Integration: Standard Functions

Questions 1–45

Question 1.

Integration of e^{kx} Find

a) $\int (e^x + x)\,dx$

b) $\int e^{2x}\,dx$

c) $\int (2e^{3x} - x^2)\,dx$

d) $\int (\sqrt{e^x} - \frac{9}{x^2})\,dx$

e) $\int \frac{2 + e^x}{e^{8x}}\,dx$

Question 2. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = e^{-3x}$

b) $\frac{dy}{dx} = 4 - 2e^{-2x}$

c) $\frac{dy}{dx} = \sqrt{x} - \frac{1}{3}e^{x/6}$

d) $\frac{dy}{dx} = x - \frac{6}{e^{4x}}$

Question 3. The curve y has gradient function $\dfrac{dy}{dx} = 12e^{6x}$ and passes through the point $(\ln 2, 128)$. Show that y can be written in the form $y = ae^{bx} + c$, where a, b and c are integers.

Question 4. The curve y has gradient function $\dfrac{dy}{dx} = \frac{24}{\sqrt{e^{3x}}}$ and passes through the point $(\ln 4, 1)$. Show that y can be written in the form $y = a + be^{cx}$, where a and b are integers and c is a rational number.

Question 5. Evaluate the following definite integrals:

a) $\int_0^3 (e^x + 2)dx$

b) $\int_{\ln 2}^{\ln 3} (100 - e^x)dx$

c) $\int_{-1}^{1} (24e^{2x/3} - 2x)dx$

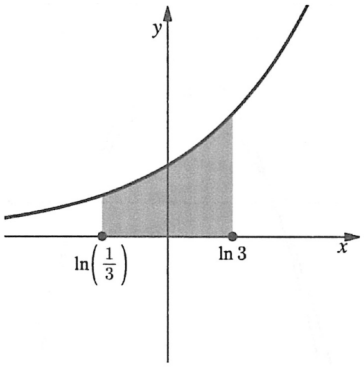
d) $\int_0^{\ln 2} (\frac{5}{e^x} - \frac{2}{e^{2x}})dx$

Question 6. Find the exact value of:

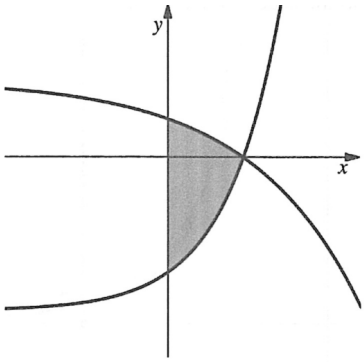
a) $\lim_{\delta x \to 0} \sum_{x=0}^{1} 3e^{18x}\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=2}^{\ln 8} \frac{1}{\sqrt[3]{e^x}}\delta x$

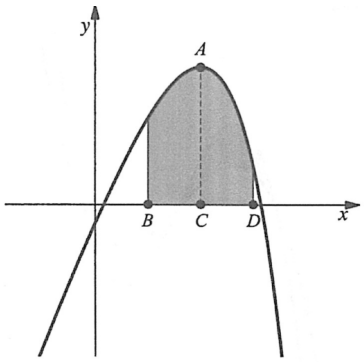
Question 7. The diagram below shows the curve with equation $y = \sqrt{e^x}$. Find the exact value of the shaded area.



Question 8. The diagram below shows the curves $y = 2 - e^x$ and $y = e^{2x} - 4$. Find the exact shaded area between the two curves.



Question 9. The diagram below shows the curve with equation $y = e^2x - e^x$. The curve has a turning point at A, and AC bisects BD. It is given that $BD = 2$ units. Find the exact shaded area.



Question 10. Integration of $\sin kx$ **and** $\cos kx$ Find

a) $\int (\sin x + \cos x)\, dx$

b) $\int (2\sin x - \cos x)\, dx$

c) $\int (2\sin 2x + \cos 4x)\, dx$

d) $\int (3\sin \frac{1}{2}x - 6\cos \frac{2}{3}x)\, dx$

Question 11. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = 3\cos 6x$

b) $\dfrac{dy}{dx} = \frac{3}{x^2} - 4\sin\frac{1}{4}x$

c) $\dfrac{dy}{dx} = \sqrt[3]{x} - \frac{2}{3}\cos\frac{5}{6}x$

Question 12. The curve y has gradient function $\dfrac{dy}{dx} = -4\sin 3x$ and passes through the point $(\frac{\pi}{6}, 3)$. Show that y can be written in the form $y = a\cos bx + c$, where b and c are integers, and a is a rational number.

Question 13. The curve y has gradient function $\dfrac{dy}{dx} = 2\sin\frac{1}{3}x - 3\cos\frac{1}{3}x$ and passes through the point $(\pi, -\frac{9\sqrt{3}}{2} - 3)$. Show that y can be written in the form $y = a\cos\frac{x}{b} + c\sin\frac{x}{b}$, where a, b and c are integers.

Question 14. Evaluate the following definite integrals:

a) $\displaystyle\int_0^{\pi/2} \sin x\, dx$

b) $\displaystyle\int_0^{\pi/4} \cos 2x\, dx$

c) $\displaystyle\int_{-5\pi/6}^{5\pi/3} 3\sin\frac{1}{5}x\, dx$

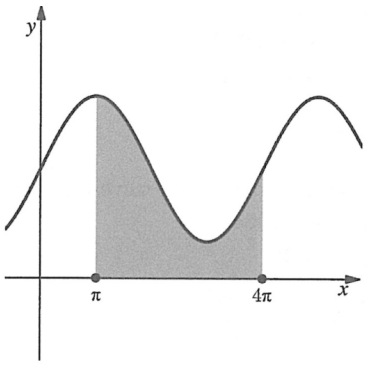
d) $\displaystyle\int_{\pi/8}^{\pi/4} (5\sin 4x - 6\cos 4x)dx$

Question 15. Find the exact value of:

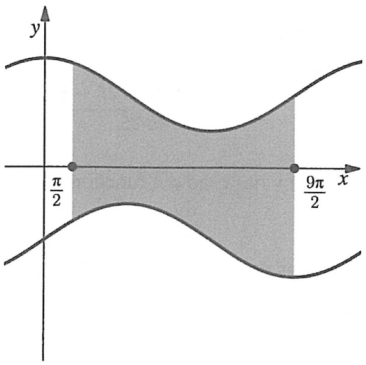
a) $\lim_{\delta x \to 0} \sum_{x=0}^{\pi/12} (8 \sin 4x - 16 \cos 4x)\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=9\pi}^{10\pi} (\frac{1}{8} \sin \frac{1}{4}x - \frac{1}{16} \cos \frac{1}{4}x)\delta x$

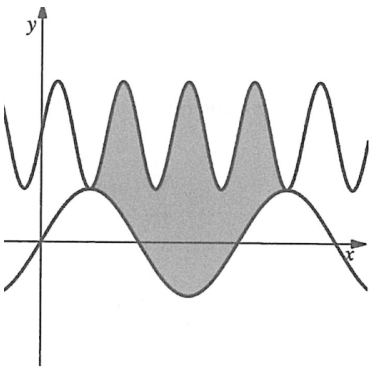
Question 16. The diagram below shows the curve with equation $y = 3 + 2 \sin \frac{1}{2}x$. Find the exact value of the shaded area.



Question 17. The diagram below shows the curves $y = \sin \frac{1}{3}x - 2$ and $y = \cos \frac{1}{3}x + 2$. Find the exact shaded area between the two curves.



Question 18. The diagram below shows the curve with equation $y = \sin x$ and the curve with equation $y = a + \sin bx$, where x is in radians. Find the value of a and the value of b, and hence find the exact shaded area.



Question 19. Integration of a^x Find

a) $\int 3^x \, dx$

b) $\int (3^{2x} + 4^{3x}) \, dx$

c) $\int (\frac{1}{5^x} - \frac{2}{2^x}) \, dx$

d) $\int \frac{3^x - 4}{9^x} \, dx$

Question 20. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = 8^x$

b) $\frac{dy}{dx} = x - 3^x \ln 3$

c) $\frac{dy}{dx} = 6^{3x} - \frac{1}{6^{3x}}$

d) $\frac{dy}{dx} = \frac{16^x - 4}{2^x}$

Question 21. The curve y has gradient function $\dfrac{dy}{dx} = 25^x$ and passes through the point $(0, 20)$. Show that $y = \dfrac{25^x + 20\ln 25 - 1}{\ln 25}$.

Question 22. The curve y has gradient function $\dfrac{dy}{dx} = \frac{1}{4^x}$ and passes through the point $(0, \frac{1}{\ln 4})$. Show that $y = \dfrac{1 - 2^{-2x-1}}{\ln 2}$.

Question 23. Evaluate the following definite integrals:

a) $\int_{-1}^{1} 3^x dx$

b) $\int_{0}^{2} (2^x - 4^x) dx$

c) $\int_{-1/3}^{1/3} (5^{3x} - x) dx$

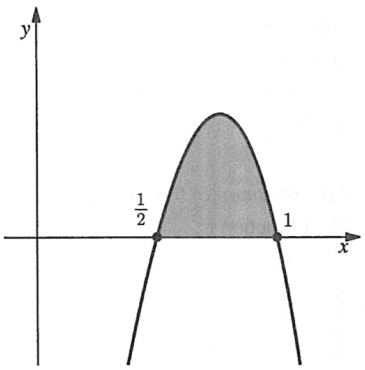
d) $\int_{1}^{2} \frac{2 - 4^x}{4^x} dx$

Question 24. Find the exact value of:

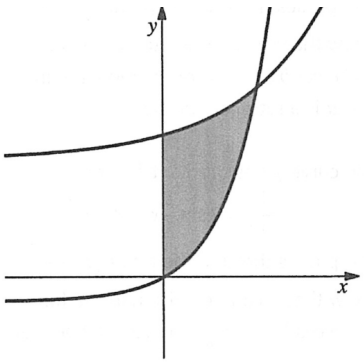
a) $\lim_{\delta x \to 0} \sum_{x=5}^{10} 5^{0.2x} \delta x$

b) $\lim_{\delta x \to 0} \sum_{x=1/10}^{1/5} \frac{10}{2^{5x}} \delta x$

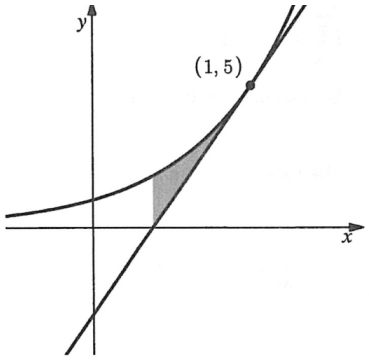
Question 25. The diagram below shows the curve with equation $y = 4x - 4^x$. Find the exact value of the shaded area.



Question 26. The diagram below shows the curves $y = 2^x + 5$ and $y = 4^x - 1$. Find the exact shaded area between the two curves.



Question 27. The diagram below shows the curve with equation $y = 5^x$. The tangent to the curve at $(1, 5)$ is also shown. Find the exact shaded area.



Question 28. Integration of $\frac{1}{x}$ Find

a) $\int \frac{1}{x}\,dx$

b) $\int (\frac{3}{x} - \frac{x}{3})\,dx$

c) $\int (\frac{5}{3x} + \frac{2}{5x^2})\,dx$

d) $\int \frac{3+8x}{4x}\,dx$

Question 29. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{4}{x} - \frac{4}{x^2}$

b) $\frac{dy}{dx} = 2 - \frac{7}{4x}$

c) $\frac{dy}{dx} = \frac{3\sqrt{x}+1}{x^{3/2}}$

d) $\frac{dy}{dx} = \frac{8x-3}{5x^2}$

Question 30. The curve y has gradient function $\frac{dy}{dx} = \frac{5}{x}$ and passes through the point $(e^5, 1)$. Show that y can be written in the form $y = a + b\ln|x|$, where a and b are integers.

Question 31. The curve y has gradient function $\frac{dy}{dx} = \frac{3-10x}{x}$ and passes through the point $(1, 1)$. Show that y can be written in the form $y = ax + b + c\ln|x|$, where a, b and c are integers.

Question 32. Evaluate the following definite integrals:

a) $\int_1^2 \frac{9}{x} dx$

b) $\int_{-3}^{-1} \frac{8}{x} dx$

c) $\int_1^4 \frac{2+x}{10x} dx$

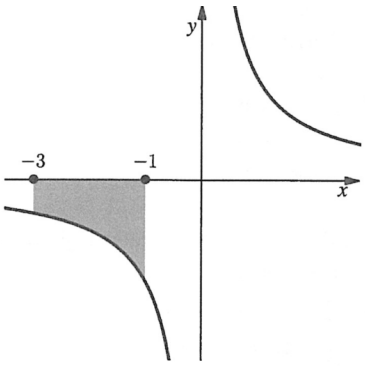
d) $\int_{-3}^{-2} \frac{3-x}{5x} dx$

Question 33. Find the exact value of:

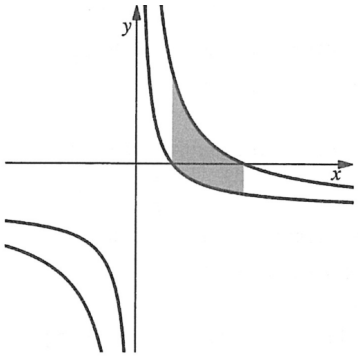
a) $\lim_{\delta x \to 0} \sum_{x=-8}^{-4} (\frac{4}{x} - 1)\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=e}^{e^2} (\frac{3}{2x} + 12x)\delta x$

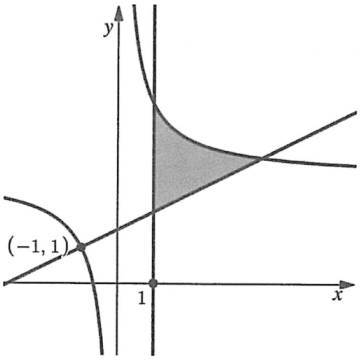
Question 34. The diagram below shows the curve with equation $y = \frac{10}{x}$. Find the exact value of the shaded area.



Question 35. The diagram below shows the curves $y = \dfrac{9-3x}{x}$ and $y = \dfrac{3-3x}{x}$. Find the exact shaded area between the two curves.



Question 36. The diagram below shows the curve with equation $y = 3 + \frac{2}{x}$. The normal to the curve at $(-1, 1)$ is also shown. Find the exact shaded area, bounded by the curve, the normal and the line $x = 1$.



Question 37. Integration of Trigonometric Functions Find

a) $\int \sec^2 x \, dx$

b) $\int \operatorname{cosec} x \cot x \, dx$

c) $\int \sec x \tan x \, dx$

d) $\int \operatorname{cosec}^2 x \, dx$

e) $\int \sec^2 4x \, dx$

f) $\int \operatorname{cosec}^2 \frac{1}{3}x \, dx$

Question 38. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \operatorname{cosec} 2x \cot 2x$

b) $\frac{dy}{dx} = 8 \operatorname{cosec} \frac{1}{4}x \cot \frac{1}{4}x$

c) $\frac{dy}{dx} = \frac{2}{3} \sec \frac{10}{3}x \tan \frac{10}{3}x$

Question 39. The curve y has gradient function $\dfrac{dy}{dx} = 5\sec^2 3x$ and passes through the point $(\frac{\pi}{12}, 5)$. Write y in terms of x.

Question 40. The curve y has gradient function $\dfrac{dy}{dx} = \sec \frac{1}{2}x \tan \frac{1}{2}x$ and passes through the point $(\frac{\pi}{3}, 0)$. Write y in terms of x.

Question 41. Evaluate the following definite integrals:

a) $\displaystyle\int_0^{\pi/6} 4\sec^2 2x\,dx$

b) $\displaystyle\int_{\pi/2}^{3\pi/2} 9\operatorname{cosec}\frac{1}{3}x\cot\frac{1}{3}x\,dx$

c) $\displaystyle\int_{\pi/6}^{\pi/4} \frac{1}{3}\operatorname{cosec}^2 3x\,dx$

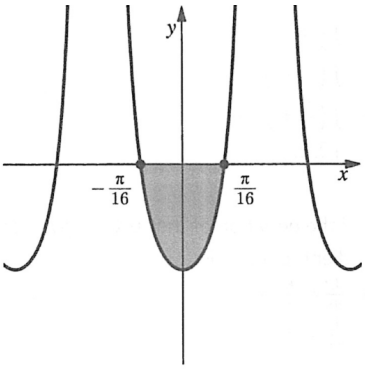
d) $\displaystyle\int_{-\pi/4}^{0} -\sec\frac{2}{3}x\tan\frac{2}{3}x\,dx$

Question 42. Find the exact value of:

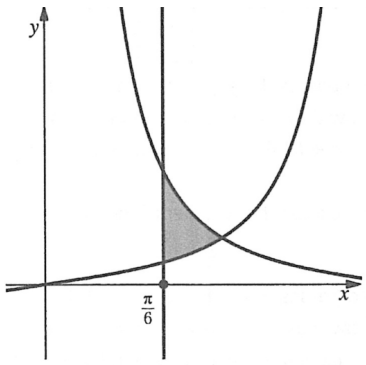
a) $\displaystyle\lim_{\delta x\to 0}\sum_{x=-\pi/2}^{\pi/2}\frac{5}{4}\sec^2\frac{5}{6}x\,\delta x$

b) $\displaystyle\lim_{\delta x\to 0}\sum_{x=\pi/9}^{2\pi/9}\frac{3}{8}\sec\frac{3}{2}x\tan\frac{3}{2}x\,\delta x$

Question 43. The diagram below shows the curve with equation $y = \sec^2 4x - 2$. Find the exact value of the shaded area.



Question 44. The diagram below shows the curves $y = \operatorname{cosec} x \cot x$ and $y = \sec x \tan x$. Find the exact shaded area between the two curves.



Question 45. The diagram below shows the curve with equation $y = \text{cosec}^2 x$. AB is the tangent to the curve at $x = \frac{\pi}{4}$. Find the exact shaded area, bounded by the curve, the tangent, the coordinate axes, and the line $x = \frac{\pi}{2}$.

